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FEATURE ARTICLE

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Training and learning support for people with vision impairment in the use of smartphones and applications (apps): an exploratory Australasian survey

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Abstract

Introduction: Occupational therapists working in low vision rehabilitation make recommendations for assistive technology (AT) and provide training in AT use for people with vision impairment (PVI) to support participation and independence. Smartphones and apps are an important AT for PVI, yet little is known about the training needs of PVI or what training is currently provided. Research exploring PVI's learning and training experiences is required to inform training development.

Purpose: This study aimed to explore the perspectives, needs, and recommendations of PVI from Australia and Singapore on smartphone training methods.

Methods: An online survey collected data between November 2020 to February 2021. Participants were recruited from Australia and Singapore using purposeful sampling. The survey had three sections: (i) demographics, (ii) use of smartphones and apps, and (iii) training. It consisted of 26 closed and open-ended questions. The quantitative results from the survey were analysed descriptively, and responses to open-ended questions were analysed using content analysis.

Results: Sixty-eight PVI responded to the survey, with 34 (50%) participants from each country. There were more Australians ($n = 19/34$, 55.9%) who had accessed formal training compared to Singaporeans ($n = 11/34$, 32.3%). Participants valued both formal and informal training, and self-training was the most used method for informal training (Australia: $n = 29/34$, 85%, Singapore: $n = 22/34$, 64.7%). Participants stated they preferred individualised formal training that caters to their learning needs and is provided by patient and knowledgeable trainers. They also preferred formal training, which is flexible and convenient to access, including online training with peers.

Conclusion: Findings such as providing individualised training, ensuring increased awareness of formal training, and using both formal and informal

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training methods can be considered by occupational therapists to enhance and develop training for PVI in the use of smartphones and apps.

Consumer and community consultation: Consumers were involved at the development stage. Three people with vision impairment reviewed and provided feedback on the survey's accessibility and content.

PLAIN LANGUAGE SUMMARY

This study looked at how occupational therapists can better help PVI learn to use smartphones and apps. Using a smartphone is important for PVI, but it is not clear what kind of training PVI need or what training is available. So researchers wanted to hear directly from PVI about their experiences and needs regarding smartphone training. Altogether, 68 PVI from Australia and Singapore completed a survey online. The study found that more Australians had received formal training compared to Singaporeans. Most participants were found to have trained themselves, but they wanted personalised formal training that met their needs. They liked trainers who were patient, knowledgeable, and flexible, and they valued online training with peers. The study suggests that occupational therapists should consider providing personalised training, making PVI more aware of training that does exist and using a mix of formal and informal methods to help PVI better use smartphones and apps.

KEYWORDS

blind, people with vision impairment, smartphone training

1 | INTRODUCTION

A core area of occupational therapy practice is assistive technology (AT), where occupational therapists work with clients to select and use AT that supports participation. The successful adoption of AT is mediated via a process that is client-centred and evidence-informed, involving assessment, selection and matching of AT, and identifying and providing suitable training (American Occupational Therapy Association, 2016; de Jonge et al., 2017; Thawisuk et al., 2023). A speciality area of AT is low vision rehabilitation, where occupational therapists work with people with vision impairment (PVI) to access suitable AT (McGrath & Corrado, 2018).

PVI include those who are either blind or have low vision, as defined by the International Classification of Diseases 11 (World Health Organization, 2022). In a highly visual world, PVI need AT to access information in the environment to enhance their independence (Senjam, 2021). For PVI, traditional technologies, such as closed-circuit TVs, Braille GPS, and electronic magnifiers, can be bulky and draw unnecessary attention to the user (Irvine et al., 2014), and having a specific device for each task is cumbersome and impractical and leads to higher costs and complexity for PVI (Awad et al., 2018). Smartphones, which became prevalent around 2007, have largely addressed the many drawbacks associated with traditional AT and barriers to usability. For example, with the introduction of in-built accessibility features in smartphones, such as speech-to-text and vice versa and text size adjustment (Chun et al., 2012), it became possible to replace multiple traditional devices with a smartphone (Crossland et al., 2014). Previously, PVI had to purchase additional software for traditional AT to access these features, increasing expenses and limiting their access (Martiniello et al., 2019). Besides the in-built

Key Points for Occupational Therapy

- People with vision impairment (PVI) report utilising smartphones as an effective AT device.
- Formal and informal training enhances PVI's smartphone and app use, which increases their participation.
- To better support PVI, sighted occupational therapists should consider partnering with trainers with lived experience of vision impairment.

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features of smartphones, availability of applications ('apps') at a low or no cost, designed specifically to enable PVI to overcome issues with activities of daily living and navigation in the community, also expanded exponentially (Csapo et al., 2015; Wong & Tan, 2012). Balata and Mikovec (2016) demonstrated smartphones and apps were valuable to PVI in the Czech Republic, as they enabled increased independence and better quality of life. In particular, the use of smartphones and apps allowed participants to travel to unfamiliar places more independently, as well as access printed information more readily.

While smartphones and apps provide solutions to traditional AT in everyday life, like any technology, many users need suitable training to use them optimally (Locke et al., 2021; Sankhi & Sandnes, 2020). Educating clients on the use of AT is an important part of occupational therapists practice in the provision of AT, yet it can be challenging for occupational therapists to prioritise training (Thawisuk et al., 2023). As a result, AT users may need to train themselves in the use of their technology. Martiniello et al. (2019) reported that the majority of their 466 international participants (69%) with vision impairment surveyed relied on self-training to use their new smartphone and only 7.5% had received training from vision rehabilitation professionals. Similar results were found in a survey of 33 PVI in the USA exploring the use of accessibility features of iPhones and iPads; 63% of participants relied on self-training, 12% received training from an eye care professional, 12% learnt from the Apple store associate or equivalent, and the remaining 12% used a combination of these methods (Robinson et al., 2017). Though self-training appears to be the most common method in learning to use smartphones and apps by PVI, it is unclear whether this method is an active choice or simply the only method available. In Locke et al. (2021), 65% of participants reported that they would like to see more training being provided by their vision support organisation. Sankhi and Sandnes (2020) were also in support of more training resources, stating that smartphone training materials should be made easily available to schools to support youths with vision impairment. Even though training in smartphone use is highlighted to be essential for PVI, there are few papers examining it.

To date, few studies have been published that investigate smartphone training programmes for PVI. Miskin et al. (2021) found that for all 337 veterans (100%) with vision impairment, their proficiency in the use of smartphones and tablets improved after a 35-hour professional training package and they needed 'less' or 'no help' in the use of smartphones post-training. A qualitative study in Norway (Fuglerud et al., 2021) explored solutions and barriers to the use and mastery of smartphones by older PVI. They found that improving the provision of training and organising good post-training support would better reach and motivate older PVI to learn to use smartphones and apps. In earlier research by Fuglerud et al. (2018), which explored training older people to use a mainstream smartphone with an EziSmart keypad and apps which the team developed, training and post training follow-up was found to contribute to learning, confirming the importance of training to use smartphones effectively.

Previous research has found that smartphones are becoming a core piece of AT for PVI, and a small number of studies have indicated the value of professional training to learn these new technologies. A gap in the literature remains in the area of understanding smartphone training and learning needs of PVI, to inform the development and delivery of such training. Both Australia and Singapore have organisations providing formal training in the use of smartphones for PVI, where occupational therapists are among the professionals who devise and carry out the training to enable participation (Tan et al., 2023). Thus, in this study, we aimed to understand the training and support needs of PVI and current provision of training in Australasia, with the aim to identify

- patterns of use of smartphones and apps by PVI;
- different training methods PVI used;
- PVI's perspectives on the training they received;
- the challenges PVI encountered; and
- recommendations to inform the development and delivery of training.

2 | METHODS

2.1 | Design

A cross-sectional survey methodology guided this study. An online survey was designed, which allowed data to be collected from a purposeful sample of PVI living in Australia or Singapore. The Checklist for Reporting Results of Internet E-Surveys (CHERRIES) guided the reporting of this study (Eysenbach, 2004).

2.1.1 | Ethics

Ethical approval was obtained from the Institutional Review Board from the Singapore Institute of Technology (Project number 2020150), as well as the Human Research Ethics Committee from the University of Queensland (Approval number 2020002092). Participants consented to enter the survey by checking the consent

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box after reading the information sheet. For participants who could not respond to the survey online, the option of participating via a phone call was provided. For three participants who participated via phone calls, consent was obtained verbally at the beginning of the phone call.

2.2 | Positionality

The researchers HLT, TA, HG, and TM are occupational therapists without vision impairment but possess extensive experience in disability sector research. AS has a research background in public health. The research was conducted within the scope of HLT's PhD programme. HLT's volunteer work with a Singaporean service for PVI motivated the research. The researchers' positionality reflects their professional and personal ties within the disability sector, which may affect how they view and understand the study findings.

2.3 | Survey development and pre-testing

The 'Learning to Use Smartphone and Apps' survey was developed based on information from literature and questions adapted from the 'Technology Utilisation Survey 2020' from CNIB Foundation in Canada. The 26 questions centred around participants' use of smartphones and apps as well as their perspectives on the different methods they used to learn to use smartphones and apps optimally. Section 1 of the survey consisted of a demographic questionnaire to collect basic information about participants' type of eye pathology, the impact of vision impairment on daily activities, age, and occupation. Section 2 of the survey aimed to explore participants' use of smartphones and apps, and section 3 specifically explored issues surrounding training and learning support. A combination of closed questions with Likert scale responses or multi-choices and two open-ended questions were included in section 3 of the survey. The multi-choice questions provided the option to elaborate if the response of 'other' was chosen. The two open-ended questions focused on seeking suggestions on how to improve the training and learning support, which they have received and asking participants to list what they would want if they could receive any kind of training and learning support in the future. The Qualtrics platform was used to develop the online survey to ensure accessibility for PVI since it has the inbuilt feature of optimising view for delivery on various devices in large font, as well as allowing for the creating of page breaks, and a visual marker of progress for ease of completion (Qualtrics, 2020). Prior to launching the survey, it was piloted with a total of three PVI in Australia and Singapore to assess the accessibility of the survey and if the questions were worded appropriately to limit ambiguity. We have included the survey as a supporting information.

2.3.1 | Recruitment and data collection

To carry out this study, seven community non-government organisations including Guide Dogs Australia, Guide Dogs Western Australia, VisAbility, Vision Australia Radio, and Guide Dogs Singapore, which support PVI directly or indirectly in Australia and Singapore, were approached to help in the recruitment of participants. The organisations distributed the electronic flyer for the survey via their various social media channels. Participants were able to click on a link on the flyer to partake in the online survey or contact the first author to complete the survey via telephone. The flyer indicated that to be eligible to participate, participants needed to have any level of vision impairment, be above 18 years old and be able to complete the survey in English. A total of 97 participants commenced the survey. Among them, one did not meet one of the inclusion criteria, and 28 were excluded as they completed less than 60% of the questions. This resulted in 68 participants, who completed between 60% and 100% of the questions. On average, the participants completed 92% of the questions.

2.3.2 | Data analysis

Data extraction from Qualtrics involved using descriptive statistics to describe the quantitative results via the software IBM® *SPSS*® Statistics 26. Responses to the two open-ended questions that asked participants about suggestions to improve training and preferences for future learning support were analysed using content analysis with themes derived by grouping similar responses together by the first author (Portney, 2020).

3 | RESULTS

3.1 | Participants' characteristics

The demographic characteristics of the 68 ($n = 34$ Australian; $n = 34$ Singaporean) participants are displayed in Table 1. The mean age of the Australian participants (54.8 years) was higher compared to the Singaporean participants (42.4 years). Overall, they comprised a highly educated group, with more than half of the participants from both Australia ($n = 24/34$, 70.6%) and Singapore ($n = 21/34$, 61.8%) having a post-

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TABLE 1 Demographics and extent of vision impairment of participants.

Demographic	Australian ($N = 34$)	Singaporean ($N = 34$)	Total
Gender (*N*/%)			
Male	13 (38.2)	18 (52.9)	31
Female	21 (61.8)	15 (44.1)	36
Prefer not to say	0	1 (2.9)	1
Highest level of qualification (*N*/%)			
Primary school (year 10 or equivalent)	0	1 (2.9)	1
Secondary school (year 12 or equivalent)[a]	10 (29.4)	12 (35.3)	22
Diploma/certificate	8 (23.5)	6 (17.7)	14
Bachelor's degree	10 (29.4)	6 (17.7)	16
Postgraduate degree or beyond	6 (17.7)	9 (26.5)	13
Employment status (count) (*N*/%)[b]			
Full-time	3 (8.8)	18 (52.9)	21
Part-time	8 (23.5)	5 (14.7)	12
Casually	3 (8.8)	2 (5.9)	5
Unemployed (looking for work)	5 (14.71)	5 (14.7)	10
Self-employed	0	2 (5.88)	2
Not working (disability pension)	21 (61.8)	0	21
Retired (government aged care pension)	4 (11.8)	0	4
			6
Retired (self-funded)	1 (2.9)	5 (14.7)	4
Volunteer	4 (11.8)	0	
Gross monthly household income (*N*/%)[c]			
Below 1,000	2 (5.9)	7 (20.6)	9
1,000–2,000	5 (14.7)	7 (20.6)	12
2,000–5,000	16 (47.1)	8 (23.5)	24
5,000–10,000	1 (2.9)	10 (29.4)	11
More than 10,000	7 (20.6)	1 (2.9)	8
Did not state	3 (8.8)	1 (2.9)	4
Severity of vision loss among participants (*N*/%)			
I have been blind/partially sighted all my life	19 (55.9)	15 (44.1)	34
I began losing my sight in childhood, <18 years	6 (17.65)	7 (20.6)	13
I began losing my sight between 18 and 55 years	7 (20.6)	9 (26.5)	16
I began losing my sight after 55 years	2 (5.9)	3 (8.8)	5
Current state of sight loss (*N*/%)			
Mild—I have most of my sight	0	3 (8.8)	3
Moderate—I have some of my sight	6 (17.7)	15 (44.1)	21
Significant—I can see very little	13 (38.2)	8 (23.5)	21
Total—I am completely blind	15 (44.1)	6 (17.7)	21
Did not state	0	2 (5.9)	2

[a]Equivalent to Singapore's GCE-N-Level, CGE-O-Level, and GCE-A-level.
[b]Participants could select more than one choice; therefore, the total does not add up to 100%.
[c]Income categories are both in AUD or SGD, which were similar in value at the point of data collection.

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secondary school education. There were significantly more Singaporean participants who worked full time ($n = 18/34$, 52.9%) than Australian participants ($n = 3/34$, 8.8%), and Singaporean participants employed full time worked longer hours (up to 50 h per week) compared to the Australian participants (up to 40 h per week). There were 21 Australian participants who indicated they were not working and receiving a government pension and four who had retired and were receiving an aged care pension. Conversely, there are no such disability support or aged pension schemes available in Singapore, which could imply the need for PVI living in Singapore to continue working in order to support themselves.

For both countries, most participants had been blind or partially sighted all their lives (Australia: $n = 19/34$, 55.9%; Singapore: $n = 15/34$, 44.1%). Some Singaporean participants ($n = 15/34$, 44.1%) had moderate vision impairment, while some of the Australian participants ($n = 15/34$, 44.1%) were completely blind.

3.2 | Use of smartphones and apps

Almost all participants paid for their smartphones, with only two participants from Singapore and three from Australia reporting they were supported with government funding to purchase a phone. Most participants (70.6%) across both countries used iPhones (for Australian: $n = 24/34$; for Singaporean: $n = 24/34$) with the remaining using Android smartphones. When asked to rate their level of confidence in using smartphones in their daily life, 64.7% ($n = 21/34$) of Australian participants and 61.7% ($n = 24/34$) of Singaporean participants indicated the highest level of confidence (Level 10) on the Likert scale.

Table 2 describes the patterns of use and participants' perception of their skill level in using smartphones, as well as how helpful smartphones and apps are to them. Most participants reported using their smartphones for 2 to 6 h daily (Australia: $n = 15/34$, 44.1%; Singapore: $n = 16/34$, 47.1%). In terms of skill level in using smartphones and apps, more Australian participants rated themselves as 'advanced' ($n = 17/34$, 50%) compared to Singaporean participants ($n = 11/34$, 32.5%). However, slightly more Singaporean participants rated themselves as 'expert' ($n = 5/34$, 14.71%) with only one Australian rating at an expert level. On a scale of 1–10 (with 10 being *extremely helpful*), 47% ($n = 16/34$) of Australian participants and 58.7% ($n = 19/34$) of Singaporean participants chose level 7 and above, indicating they found their smartphone and apps to be *very helpful* to *extremely helpful*, as shown in Figure 1.

3.3 | Methods of training and learning support utilised

Figure 2 depicts the methods of training and learning support utilised by participants, with some accessing training from more than one source. Most participants reported that they were 'self-taught'. The next highest method of learning was from online instructions and Podcasts. Some participants also accessed formal training from peer mentors or rehabilitation personnel from an organisation. When asked to state other methods for learning, one Australian participant shared that she hired a qualified professional to teach her while another obtained training notes from the internet and joined Zoom groups and WhatsApp groups and approached an Apple help desk to learn.

3.4 | Characteristics of formal training received

Of the participants who indicated that they received formal training (Australia $n = 19$; Singapore $n = 11$), some reported they were alerted to formal training external to the organisation they receive regular services from (Australia $n = 5$; Singapore, $n = 3$) or that their organisation provided it to them in house (Australia $n = 7$; Singapore, $n = 3$). In both countries, most participants reported that they did not have to pay for the training (Australia $n = 6$; Singapore $n = 5$).

Of those Australian participants who had formal training and responded to further questions about the

TABLE 2 Daily smartphone use and perception of skill level.

	Australian ($N = 34$)[a]	Singaporean ($N = 33$)
Length of smartphone and apps use in a day (frequency/%)		
<2 h	7 (20.6)	2 (5.9)
2–6 h	15 (44.1)	16 (47.1)
6–10 h	6 (17.7)	8 (23.5)
>10 h	2 (5.9)	5 (14.7)
Unsure	4 (11.8)	2 (5.9)
	Australian ($N = 34$)	**Singaporean ($N = 31$)**
Skill level in smartphones (frequency/%)		
Beginner	5 (14.7)	1 (2.9)
Intermediate	11 (32.4)	14 (41.2)
Advanced	17 (50.0)	11 (32.4)
Expert	1 (2.9)	5 (14.7)

[a]Total number of respondents.

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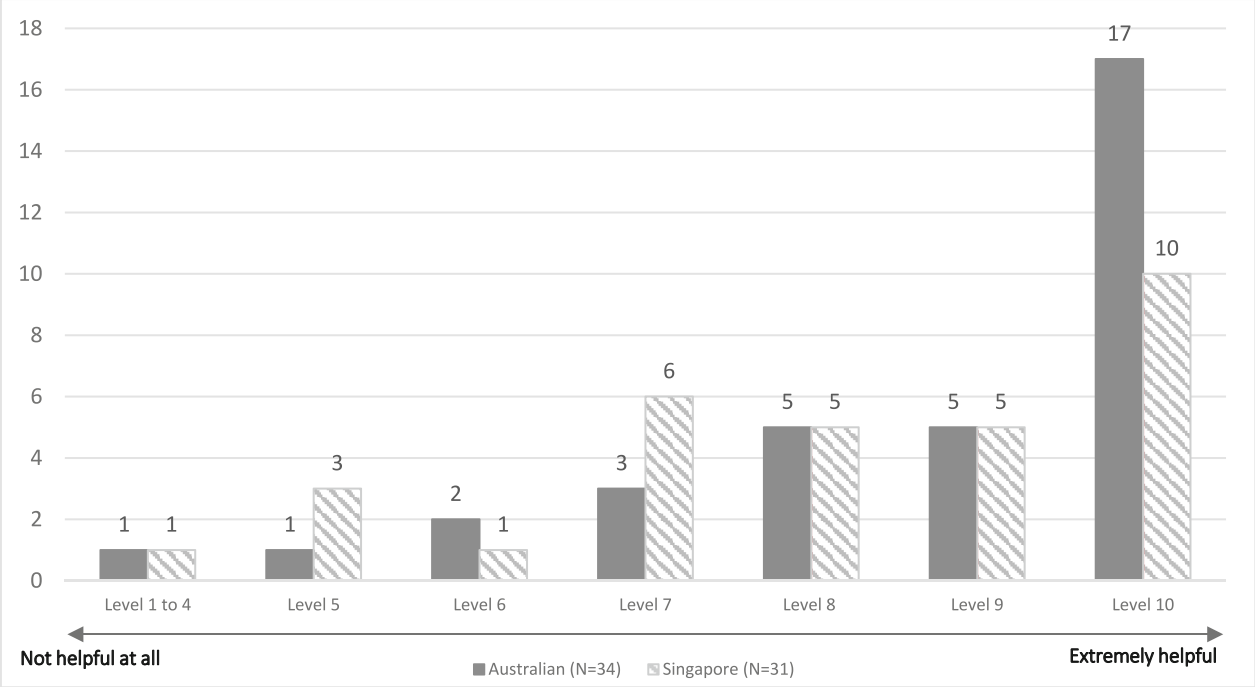

FIGURE 1 Perception of how helpful smartphones and apps are in supporting daily life.

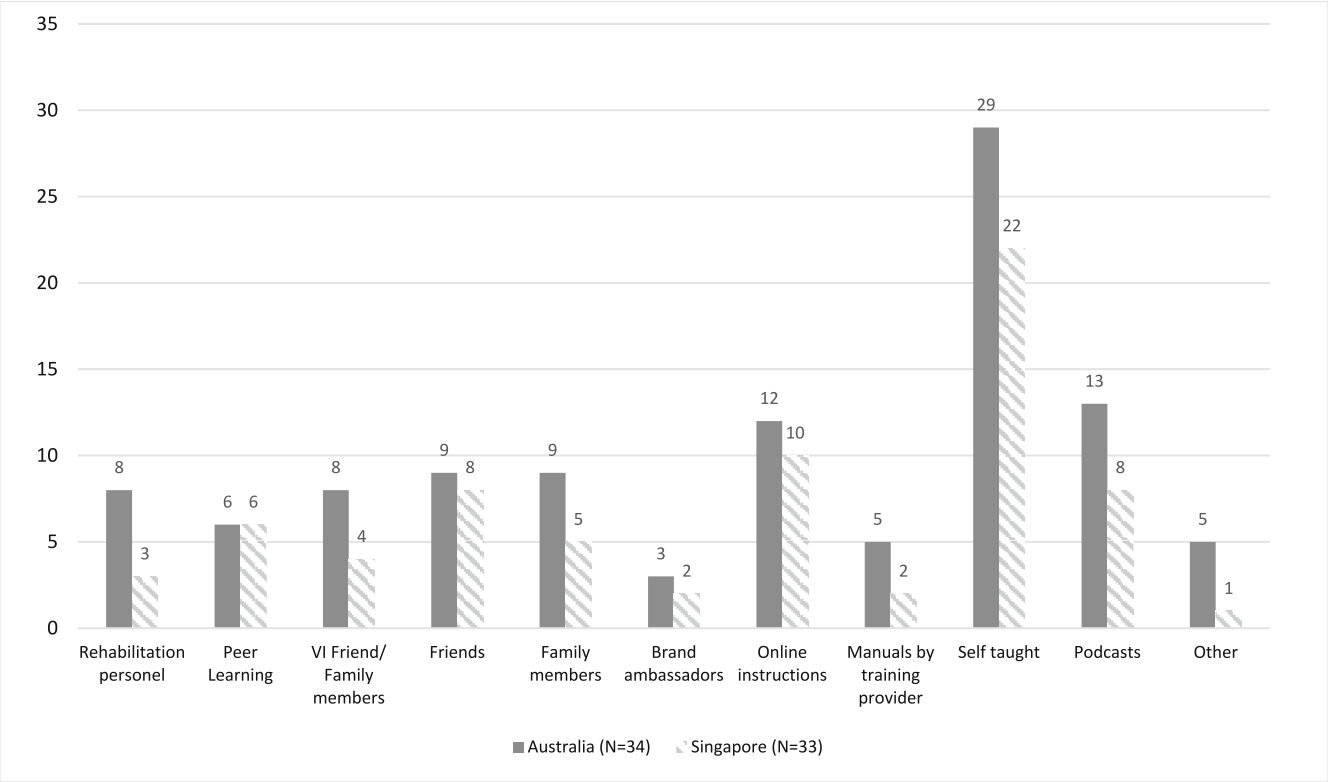

FIGURE 2 Frequencies of training methods used: Peer learning, from peer learning mentor who has vision impairment engaged by an organisation; VI, visually impaired; brand ambassadors, example, iPhone apple shop assistants; online instructions, for example, YouTube videos; manuals by training provider, from manuals developed by rehabilitation/training provider in an accessible format.

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training received, most accessed the training within 1 month of when they first had access to a smartphone ($n = 4/10$, 40%). In comparison, Singaporean participants were able to access training within a week ($n = 4/7$, 57.1%) of having their smartphone. More than half of the participants reported that they were satisfied with the time frame in which they received training (Australia: $n = 6/10$, 60%; Singapore $n = 4/7$, 57.1%). The most common method of formal training was face-to-face individual sessions (Australia: $n = 7/10$, 70%; Singapore $n = 5/8$, 62.5%). For both countries, more than half of the participants received two to four face-to-face, individual training sessions (Australia $n = 4/7$, 57.1%; Singapore $n = 3/5$, 60%). These sessions tended to be between 60 and 90 min long for Singaporean participants ($n = 3/5$, 60%) and up to 2 h for Australian participants ($n = 3/7$, 42.9%).

3.5 | Content learnt in formal and informal training

Among those who indicated that they had received formal training from organisations, the most commonly learnt skills for Australian participants were screen reader and accessibility motion gestures. For Singaporean participants, screen readers and voice assistants were most frequently learnt. Additionally, Singaporean participants mentioned under 'other skills' that they learnt foundational knowledge to navigate the phone and set up the iPhone as an assistive device. Other skills learnt by Australian participants included texting, emailing, making phone calls, and using the accessibility functions.

On the other hand, when asked to comment on what they learnt from friends, podcasts, and YouTube videos not linked to training from an organisation (informal training), some Australian participants reported that they learnt how to use apps such as a magnifier from the podcast, the camera app from their children, how to use rotor (a set of VoiceOver gestures on iOS platform), how to use generic apps, or how to set up phones and send and receive emails. Some Singaporean participants reported learning how to set up the phone; how to get the phone running or having taught themselves basic functions of the phone to meet their needs; how to use some apps like google maps and zoom; or how to use shortcuts to operate the phone.

3.6 | Follow up post formal training

The survey asked participants about the availability of follow-up post-training. For both countries, participants reported (Australia $n = 8/9$; 88.9%; Singapore $n = 4/7$, 57.14%) that they were able to call or email someone from the organisation they had training with if they needed any help post-training. One Singaporean participant commented that they received summary notes post-training via email after requesting them, and another participant from Australia commented that additional notes were provided in the CD format when elaborating answers under 'other'.

3.7 | Positive aspects of formal training

Figure 3 shows aspects of the formal training, which participants rated as positive. Among participants who received formal training and responded to this question (Australia $n = 8$; Singapore $n = 7$), all Australian and Singaporean participants rated the trainer being patient, as a positive aspect. For both countries, the two aspects that PVI found to be most useful about the training received were (i) being able to learn enough to increase their independence (Australia, 100%; Singapore, 100%) and (ii) they felt more confident after training (Australia $n = 7/8$, 87.5%; Singapore $n = 3/7$, 42.9%). When asked to comment on how useful the training had been, an Australian participant (A19) commented that attending training on how to use an iPhone made him part of the *iPhone culture* and for a Singaporean participant (S19), the training offered the 'option to be more independent [and provided] additional help to pursue independence'. A20 who participated in group training sessions stated that he liked the 'social interaction among group members', which enabled him 'to build up [a] network', while A27 commented that she liked how the sighted trainer could understand her needs despite not having a vision impairment. S19 mentioned that for her, the 'one-to-one individualised training' was what she liked about the training received.

3.8 | Negative aspects of formal training

When asked to elaborate on negative aspects of formal training received under 'other', A19 commented that 'it was difficult to access funding for training when one is over 65 years old'. A20 said that group training could not cater to individual needs well, such as the learning pace and style. For S11, the need to travel to the organisation for training was a hassle as it was 'too far and too costly to hire a taxi' for that. S19 felt that the organisation they engaged with was not 'focusing on providing training readily'.

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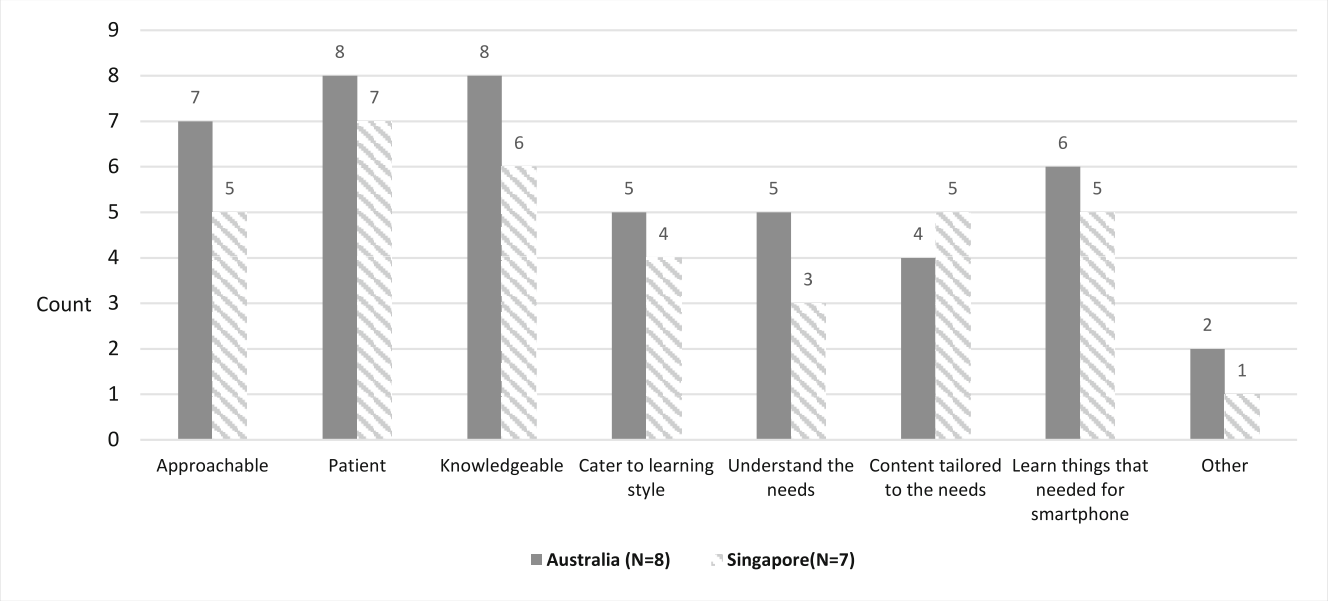

FIGURE 3 Positive aspects of formal training.

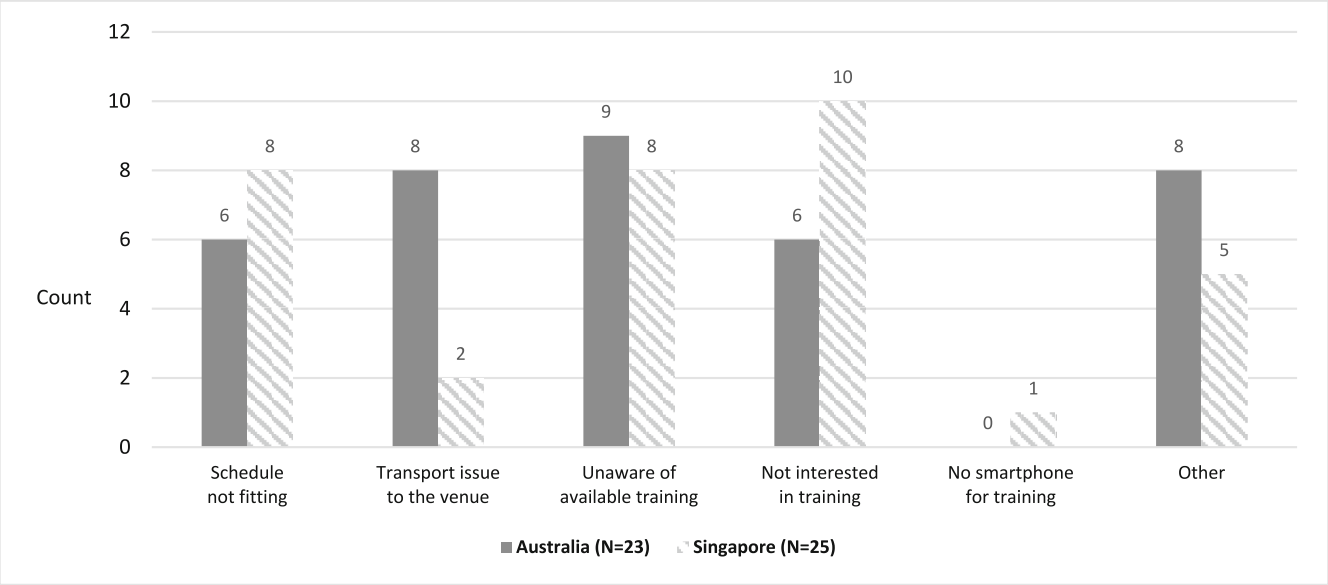

FIGURE 4 Challenges in accessing training.

3.9 | Challenges and enablers regarding accessing formal training

3.9.1 | Challenges

The survey asked participants who did not receive formal training about challenges that they encountered when attempting to access training (please see Figure 4). Singaporean participants reported the timing offered for training did not fit their schedules ($n = 8/25$, 32%) and 32% ($n = 8/25$) of participants reported that they were not made aware that training was available. Australian participants rated not being aware of the training ($n = 9/23$, 39.1%) as the biggest challenge they encountered in accessing training, followed by a lack of transport to get to the venue ($n = 8/23$, 34.8%).

When elaborating on the 'other challenges', A18 commented that a self-taught method was sufficient; hence, there was no need to access formal training, and A30 felt that the training offered was *too basic* and not targeted at his level of skills and knowledge. Online training was flagged as difficult by A32 as 'you need to own a computer and know how to use it'. For S28, having to pay for training was raised as an issue, while S32

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mentioned that training was not available when they sought it and for S31, due to her caregiving role, she had limited time for training.

3.9.2 | Enablers

The survey asked participants who did not access formal training, what factors would encourage them to utilise formal training (please see Figure 5). Australian participants reported that the biggest enabler to accessing training would be more convenient training venues or having trainers come to their homes to teach ($n = 9/19$, 47.4% for both), followed by better timing for classes ($n = 8/19$, 42.1%). A range of other factors was suggested by Australian participants including increasing *training awareness* as an enabler to access training and more *tailored courses* to meet individual needs since 'we all experience vision loss differently'. Singaporean participants reported enablers to access training would be better timing ($n = 10/23$; 43.5%), more drop-in sessions for specific issues ($n = 7/23$; 30.4%), and a range of 'other' ($n = 10/23$; 43.5%) suggestions, namely, free training but also online training sessions, 'which can be accessed any time without the need for me to travel to the venue'.

3.10 | Helpfulness of training methods

On a 5-point scale of *not at all helpful* to *very helpful* comparing formal with informal training (that included self-training, online resources such as YouTube, input from friends and family with and without vision impairment, Podcasts, and brand ambassadors), most Australian and Singaporean participants rated formal training as helpful to very helpful (Australia, $n = 16/18$, 88.9%; Singapore, $n = 6/9$, 66.7%). Self-taught, rated as the most utilised informal training method to master smartphones for participants, was reported by most participants to be *helpful* to *very helpful* (Australia, $n = 21/29$; 72.4%; Singapore, $n = 13/20$; 65%). However, some Australian participants rated it to be *somewhat helpful* to *not at all helpful* ($n = 4/29$; 13.8%) while 20% ($n = 4/20$) of Singaporean participants rated it to be *somewhat helpful*. When asked to rate training received from friends and family members who are sighted, Australian participants rated it as *somewhat helpful* ($n = 5/18$; 27.8%), while one Singaporean participant rated it as *neutral* ($n = 1/10$, 10%). Conversely, all Australian and Singaporean participants rated training from friends and family members with vision impairment to be *helpful* to *very helpful* only (Australia, $n = 8/8$, 100%; Singapore, $n = 3/3$, 100%).

3.11 | Suggestions to improve training

One main theme was derived from the qualitative responses provided by participants in both countries for suggestions on how to improve the training they received. That was for training to be made more readily available. For example, A19 suggested 'Organisation to approach government to have more trainers or have

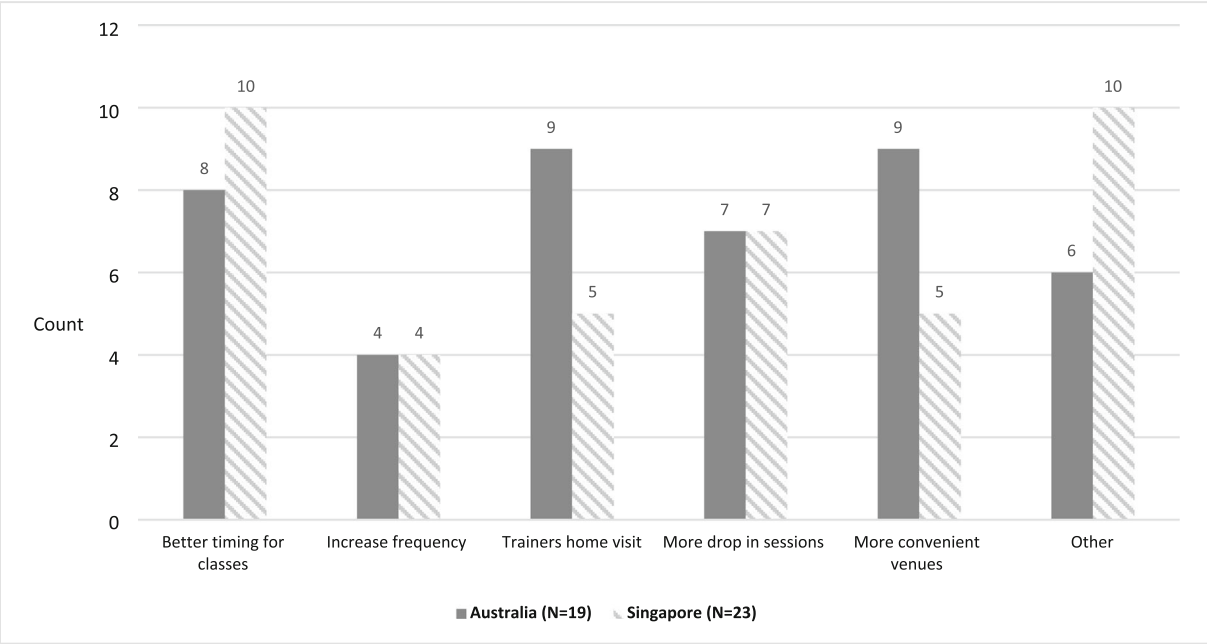

FIGURE 5 Enablers to access training.

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outside trainers which can be funded through My Aged care ... so that training can be more accessible'. S19 also stated, 'made training more available to PVI ... no increase in fees charged'.

3.12 | "Wishlist" for training and learning support

Participants were asked what would training and learning support look like if they could receive any kind of training to use their smartphones and apps. Four themes emerged when the responses were analysed. The first one was 'one-on-one training', as A2 stated '... a session in person, one to one ...', and preferably being carried out by knowledgeable yet friendly trainers with vision impairment. This point was also mentioned by S31, A2, A13, and A31. The next theme was "flexible and convenient training." This included having 'trainer come to my house', which was mentioned by most Australian participants who responded to this question (A7, A10, A31, and A32). This is likely attributed to the long distances some participants needed to travel due to the geographical characteristics of Australia as compared to Singapore. Other features identified that would make training and learning support flexible and convenient were 'better timing to suit my schedule' (S31); 'easy to access' (S21); '... more drop-in sessions at off-peak times ... training venue close to me ...' (A30), and 'online sessions with peers' (A1). Having the flexibility to go online for training was also supported by A18 and A10. The third theme that was derived was 'tailored training', which refers to training that would be customised based on the needs of learners after assessment by trainers. A7 said training should be '... in areas I want to learn, not with a trainer's own agenda ...'. S28 explained that as 'customised training for my needs ...', and A16 suggested, '... tailored training based on what I expressed ...'. The last theme was 'free training', which was included in the comments made by S21, S28, A33, and A32 as a preference of PVI.

4 | DISCUSSION

This study aimed to understand the perspectives, needs, and recommendations of PVI from Australasia about training methods they used to learn smartphones and apps. The results gathered could inform occupational therapists and other professionals supporting PVI to select and use smartphones; in particular, the findings can inform the future development and delivery of training and learning support for the optimal use of smartphones and apps. Findings show similarities and differences between the two representative countries, highlighting gaps in the provision of formal training by organisations, as well as ways to enhance informal training. The results indicate that, regardless of country, PVI use a range of methods to learn to use smartphones, with the most frequently used method being self-training. When formal training was accessed, PVI appreciated trainers who were patient, approachable, and knowledgeable. Generally, formal training was considered useful, as it increased participants' independence, confidence, and social connectivity. Participants identified travel, costs, and a lack of awareness of training availability as the main challenges to accessing formal training.

4.1 | PVI's access to training methods and their training needs

Findings from this study support earlier research that smartphones and apps are helpful AT for PVI, who use them frequently in daily life (Balata & Mikovec, 2016; Martiniello et al., 2019). The results of this study suggest that formal training is a helpful way to learn about smartphones as AT, which aligns with previous research that has found formal training to increase a sense of mastery in smartphone use among PVI (Abraham et al., 2021; Locke et al., 2021; Miskin et al., 2021; Sankhi & Sandnes, 2020; Senjam et al., 2021; Wong & Tan, 2012). However, the results also concurred with previous research that PVI are relying more on self-training, than formal training from organisations to learn to use smartphones and apps well (Martiniello et al., 2019; Robinson et al., 2017). Compared to formal training, PVI seem to have a better opportunity to learn a broader variety of content from informal methods. This opportunity may explain why many PVI choose informal training over formal training that may not cater to the individual's needs. While this suggests that informal methods may be useful to complement formal training, it may also suggest that trainers need to tailor training to individual needs more to suit different existing levels of skills and knowledge, as suggested by participants in this study. A recent study with trainers also suggests that individualised and graded approaches incorporating learners' needs are recommended, as vision impairment affects everyone differently (Tan et al., 2023). When training meets the needs and aspirations of PVI to use smartphones, it has been found to enhance participation and mitigate the negative impact of vision impairment, such as decreased quality of life, depression, and social isolation (Fuglerud et al., 2021).

The results illustrated a range of challenges that prevent PVI from accessing formal training, for example,

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excessive training fees, the inconvenience, and cost of travel to attend training, as well as a lack of awareness and promotion of available services. Lack of awareness of existing services is also an issue among health professionals (Miskin et al., 2021). This lack of awareness among health professionals leads to decreased number of referrals to appropriate services, which, in turn, results in a barrier for PVI to access smartphone training (Senjam, 2021). Along with increased knowledge and referrals by health professionals to promote formal smartphone training, enhanced co-designed marketing strategies by providers may also help to raise awareness. Participants also advocated for the greater convenience of training, including greater variability in timing, more accessible locations, having trainers conduct training at their homes, and online training options.

Online training was utilised by participants in our study and supports previous research showing online training is a widely used method to learn to use smartphones effectively (Martiniello et al., 2019). Additionally, participants used a range of training methods outside of formal and online resources, such as informal input from family and friends for a range of skills, from basic to more advanced use of the smartphone. However, like previous research (Fuglerud et al., 2018), participants in our study perceived the learning support they received from family and friends who have vision impairment to be more helpful than learning from sighted family and friends. The reliance on peers who have first-hand experience of using AT as oneself for information and support have also been found to be valuable in other AT use (Hammel et al., 2013). Due to its value, peer mentoring has also been proposed to be included in AT service delivery (Steel et al., 2016).

Despite the draw and ease of informal methods of learning, our findings indicate that learning in formal training can provide broader benefits for PVI, such as increased social participation and connection with other PVI, as well as enhanced independence. The study by Fuglerud et al. (2021) discussed that smartphone training for older PVI resulted in decreased social isolation as they were able to better connect with their friends and family after learning to use their smartphones. Increased social participation benefits one's health and wellbeing (Turcotte et al., 2018), and in particular, the negative implications of reduced social participation for older PVI have been widely discussed (Laliberte Rudman et al., 2016; Smallfield & Molitor, 2018). Therefore, when designing formal training, trainers should be cognisant of its broader benefits, such as the possibility of training being a platform to facilitate social participation.

One important element contributing to positive training experiences, which was revealed in this study, was the quality and skills of the trainer. Our findings indicate that PVI value trainers who are patient and demonstrate good listening skills during interactions and communications and that they are knowledgeable, as reported in previous research for trainers of PVI (Tan et al., 2023; Wong & Tan, 2012). Soft skills exhibited by trainers play a key part in the overall training experience, and hence, these attributes are important for trainers to consider when providing smartphone training. Occupational therapists and other professionals providing training could consider using the 'Intentional Relationship Model' (IRM), to apply the therapeutic use of self effectively when providing training (Taylor, 2020). The IRM, as a commonly used model of practice in occupational therapy, can provide trainers with the skills set and application of different communication modes for interpersonal reasoning during training, to form good rapport and be able to understand clients well (Taylor, 2020). Good communication and understanding of clients and their personal goals and values should lead to individualised training being provided.

In relation to providing formal training, the results support developing flexible and convenient training, particularly through offering online support. They would, therefore, seem to indicate either examining the myriad of online resources currently available to curate a package to be used as online support or developing new contextual and evidence-informed online training resources. Providing good online formal training can potentially mitigate the challenge of accessing formal training highlighted in the results. Developing online resources for sighted family and friends may be particularly useful given they are helping PVI to learn to use smartphones but perhaps not optimally. Online post-formal training resources may also be beneficial as post-training support has been found to be useful in supporting learning (Locke et al., 2021).

4.2 | Implications for occupational therapy practice

Training and follow-up play a crucial role in ensuring that people with visual impairments can effectively and confidently use smartphones as AT (Tan et al., 2023; Wong & Tan, 2012). The findings of this study add to the knowledge of occupational therapists regarding how smartphone training can empower PVI to use smartphones as effective AT tools for enhancing their participation and independence.

In low vision rehabilitation, occupational therapists are well suited to provide this training as experts in understanding a person's personal factors such as their

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learning capacities and interests, the myriad of environmental factors that confound or promote learning to use any smartphone as an effective AT device, and knowledge in occupational engagement to consider the use of smartphones in relation to individual routines, meaningful roles, and activities of daily living. Previous research highlights the important role that occupational therapists play in providing training and education to clients and caregivers on AT use (Kirby et al., 2020; Koh et al., 2024; McGrath et al., 2017; Persson et al., 2020). Occupational therapists are, therefore, well positioned to provide and advocate for the individualised training that PVI report wanting in this study. They are also well equipped to contribute positively to develop training programmes and resources for PVI to use smartphones, with skills in co-design and a focus on skill development tailored to individuals' needs, which were identified as a priority by PVI in this study.

It was also discussed that trainers with vision impairment are better suited to provide training to PVI given that as they share their lived experience with learners, it helps to motivate learners and alleviate some of the negative emotions expressed by learners (Tan et al., 2023). Hence, sighted occupational therapists should consider providing training in partnership with a trainer who has lived experience of vision impairment.

4.3 | Limitations and future research

Although the study had an even number of participants from both countries, its main limitation was the small sample size and thus limited response from participants who had received formal training. The results should, therefore, be considered in this context. Most participants were 40 to 55 years old; therefore, research into other age groups, particularly older adults, above 65 years old, who have been found to be a fast-growing group of PVI to use smartphones (Locke et al., 2021) is necessary. Most of the participants in this study have been blind since birth, indicating the experiences of people transiting to use smartphones from relying on vision to having to use low vision accessibility features to operate the phone requires further investigation, to explore any differences in needs. Future qualitative research is also recommended with PVI to explore 'individualised training' more in depth, to understand from PVI's perspective, the essential elements to consider when providing individualised training. Future research could focus on partnering with PVI through a co-design approach, to develop tailored training and learning support resources. This could include the development of both individualised face-to-face training and online training and learning support resources.

5 | CONCLUSION

The study aimed to explore the perspectives, needs, and recommendations of PVI from Australia and Singapore on smartphone training. PVI most commonly were self-taught, although 56% of Australians and 32% of Singaporeans did access formal training. Both formal and informal training were found to be helpful and useful by PVI. The study found that PVI want an individualised approach to formal training and value trainers' soft skills such as patience and good listening skills to better understand their needs. A range of formal training should be available from organisations as the results indicate that PVI seek more convenient and flexible formal training, including online learning, and while tailored one on one training was the preferred formal training option, group training was also considered beneficial as it promotes social connectivity. A lack of awareness of formal training is a barrier to accessing appropriate training services. Health professionals including occupational therapists and organisations that work with PVI should consider ways to effectively promote the availability of formal smartphone training for PVI.

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CONFLICT OF INTEREST

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AUTHOR CONTRIBUTIONS

Hwei Lan Tan designed the study, applied for ethical approval, carried out recruitment, data collection, analysis and interpretation, and writing the manuscript. Tammy Aplin, Tomomi McAuliffe, and Hannah Gullo were involved in study design, data analysis, and interpretation and contributed to the manuscript write-up. Ai Sun Siow was involved in data analysis and checking and formatting the manuscript.

DATA AVAILABILITY STATEMENT

Data will be available upon request.

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